

★ WEEK 4 · PART 2 of 3

Seasonality and SARIMA

Modeling the Repeating Rhythm

Seasonal differencing · $SARIMA(p,d,q)(P,D,Q)_s$ · the airline model

Part 1 tamed the trend. Part 2 adds the season — the yearly, quarterly, or weekly pattern the airline data showed — by applying the very same differencing and ARMA ideas at the seasonal lag.

2

PART 2 · SEASONALITY

Adding the Season

What seasonality is, how a seasonal difference removes it, and how SARIMA extends ARIMA with a second set of orders at the seasonal lag.

One Pattern Left to Capture

Part 1 removed the trend; the season remains

1

Differencing killed the trend, not the season

A regular difference removes a slow drift, but a yearly pattern survives — it repeats every s periods, and ordinary ARIMA cannot see it.

2

Seasonality is just memory at a longer lag

Today looks like the same month last year. That is autocorrelation at lag s — the same idea as AR/MA, moved out to the seasonal lag.

3

So we reuse everything, at lag s

Seasonal differencing, seasonal AR, seasonal MA: the Week-3 and Part-1 tools, applied a second time at the season. That is SARIMA.

SARIMA in one line: do ARIMA twice — once at lag 1, once at the seasonal lag s .

What Is Seasonality?

A pattern that repeats on a fixed calendar cycle

Seasonality is a systematic pattern that recurs every fixed number of periods s — 12 for monthly data, 4 for quarterly, 7 for daily-with-weekly rhythm.

How to spot it

- The same shape repeats every s periods, year after year.
- Peaks and troughs land in the same months each cycle.
- The ACF shows spikes at lags s , $2s$, $3s$, ...

Why it matters

- Ignoring it leaves obvious, predictable structure unmodeled.
- It shows up in residuals as leftover seasonal spikes.
- Capturing it sharply improves short-horizon forecasts.

Additive vs Multiplicative Seasonality

Do the swings stay constant, or grow with the level?

Additive

- The seasonal swing is a fixed size each cycle.
- $\text{Observed} = \text{trend} + \text{season} + \text{noise}$.
- The peaks and troughs stay the same height over time.

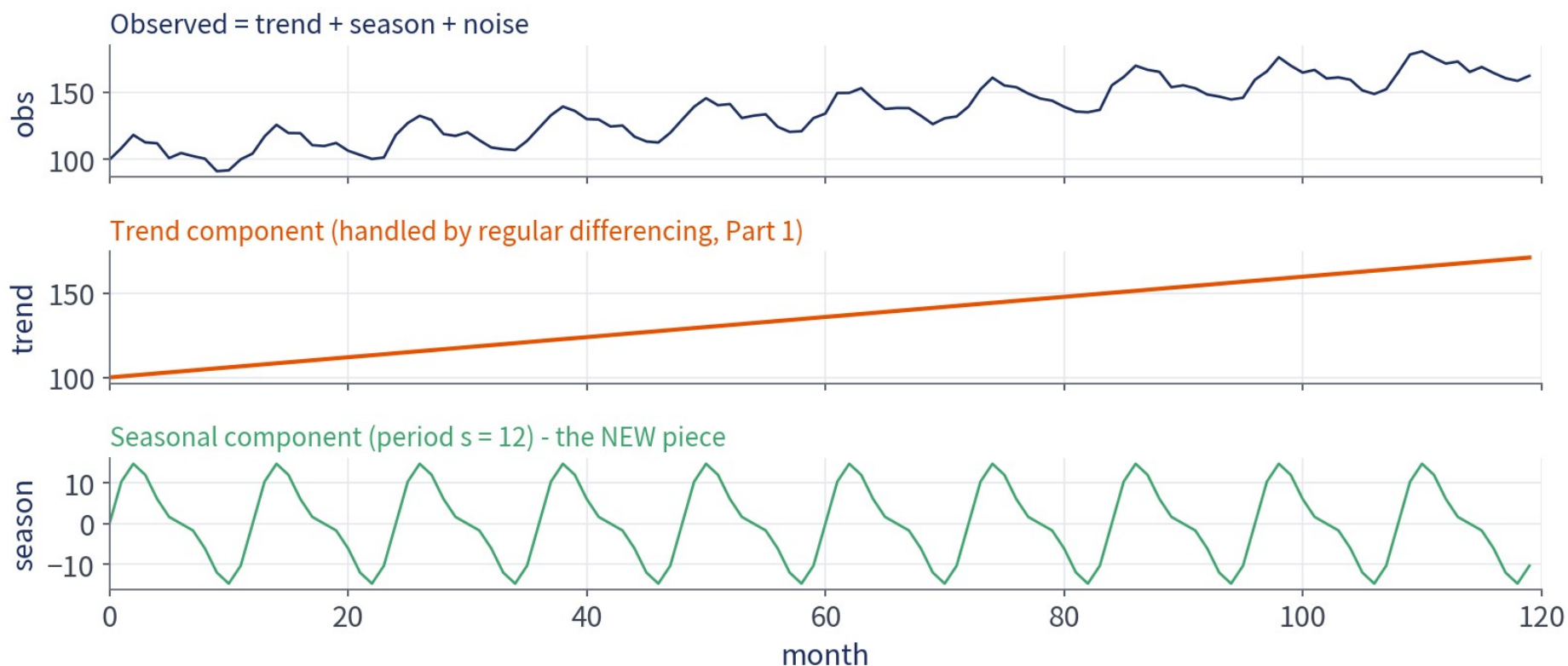
Multiplicative

- The swing scales with the level — bigger as the series grows.
- $\text{Observed} = \text{trend} \times \text{season} \times \text{noise}$ (the airline data).
- Take logs to turn it back into the additive case.

The Anatomy of a Seasonal Series

Any seasonal series splits into a trend (handled by regular differencing, Part 1), a repeating seasonal component (the new piece), and noise.

The anatomy of a seasonal series



Why Plain ARIMA Misses the Season

Adjacent-lag memory cannot reach the seasonal lag

An ordinary ARIMA models correlations at nearby lags (1, 2, 3...). A yearly pattern lives at lag 12 — too far away for a low-order model to capture without help.

The symptom

- Fit an ARIMA and the residual ACF still spikes at lag s .
- The model explains the wiggles but not the yearly rhythm.
- You could raise the order to 12+, but that is wasteful.

The fix

- Add terms that act directly at the seasonal lag s .
- A seasonal difference removes the repeating pattern outright.
- Seasonal AR/MA terms mop up what remains — enter SARIMA.

Seasonal Differencing

Subtract the value from one season ago

$$\nabla_s X_t = (1 - L^s) X_t = X_t - X_{t-s} \quad (\text{seasonal difference})$$

What it does

- Replaces each value with its change from s periods earlier.
- For monthly data, $(1 - L^{12})$ subtracts the same month last year.
- One pass removes a stable, repeating seasonal pattern.

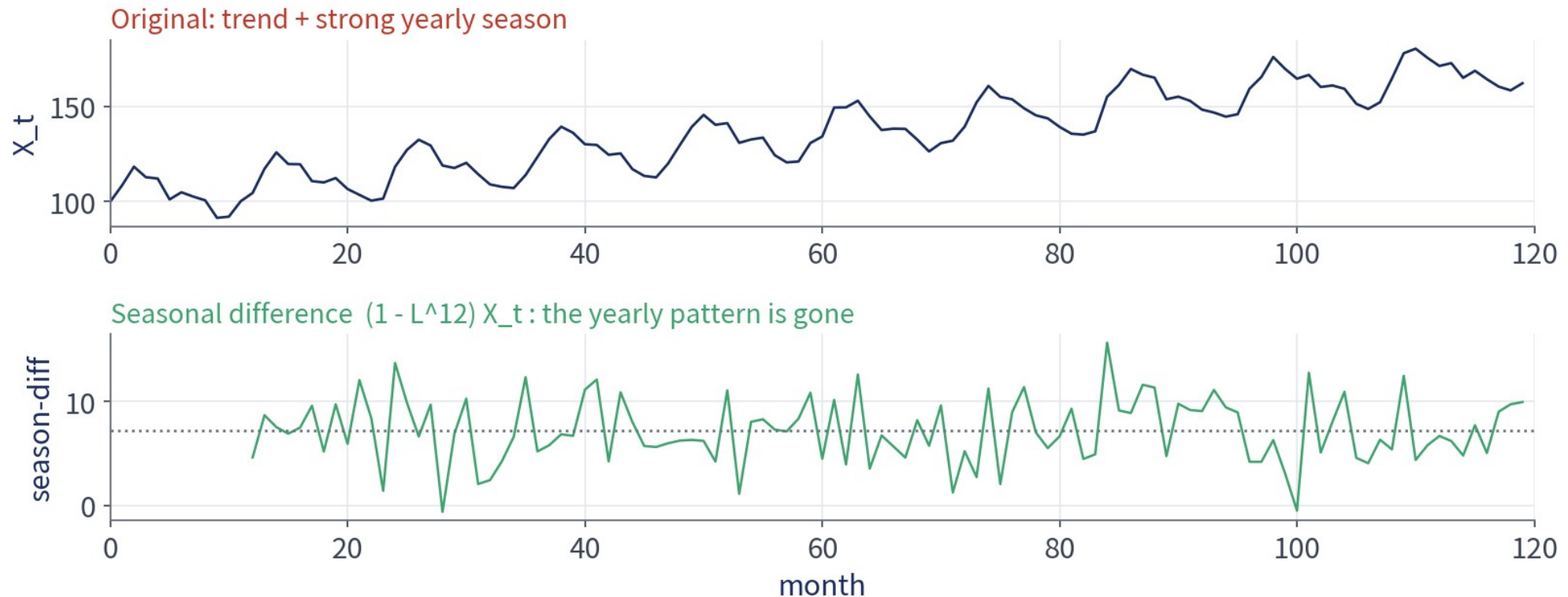
Regular vs seasonal

- A regular difference $(1 - L)$ targets the trend.
- A seasonal difference $(1 - L^s)$ targets the season.
- Trending, seasonal data often needs BOTH.

Seasonal Differencing in Action

Top: a series with a strong yearly pattern. Bottom: after $(1 - L^{12})$, the repeating season is gone and the series fluctuates around a flat mean.

Seasonal differencing subtracts the value from one season ago



The Seasonal ACF/PACF Signature

Read the seasonal lags exactly as you read the near ones

Seasonality announces itself as spikes at the seasonal lags — s , $2s$, $3s$ — in the ACF and PACF. You identify the seasonal orders (P , Q) from those spikes, just like p and q .

Where to look

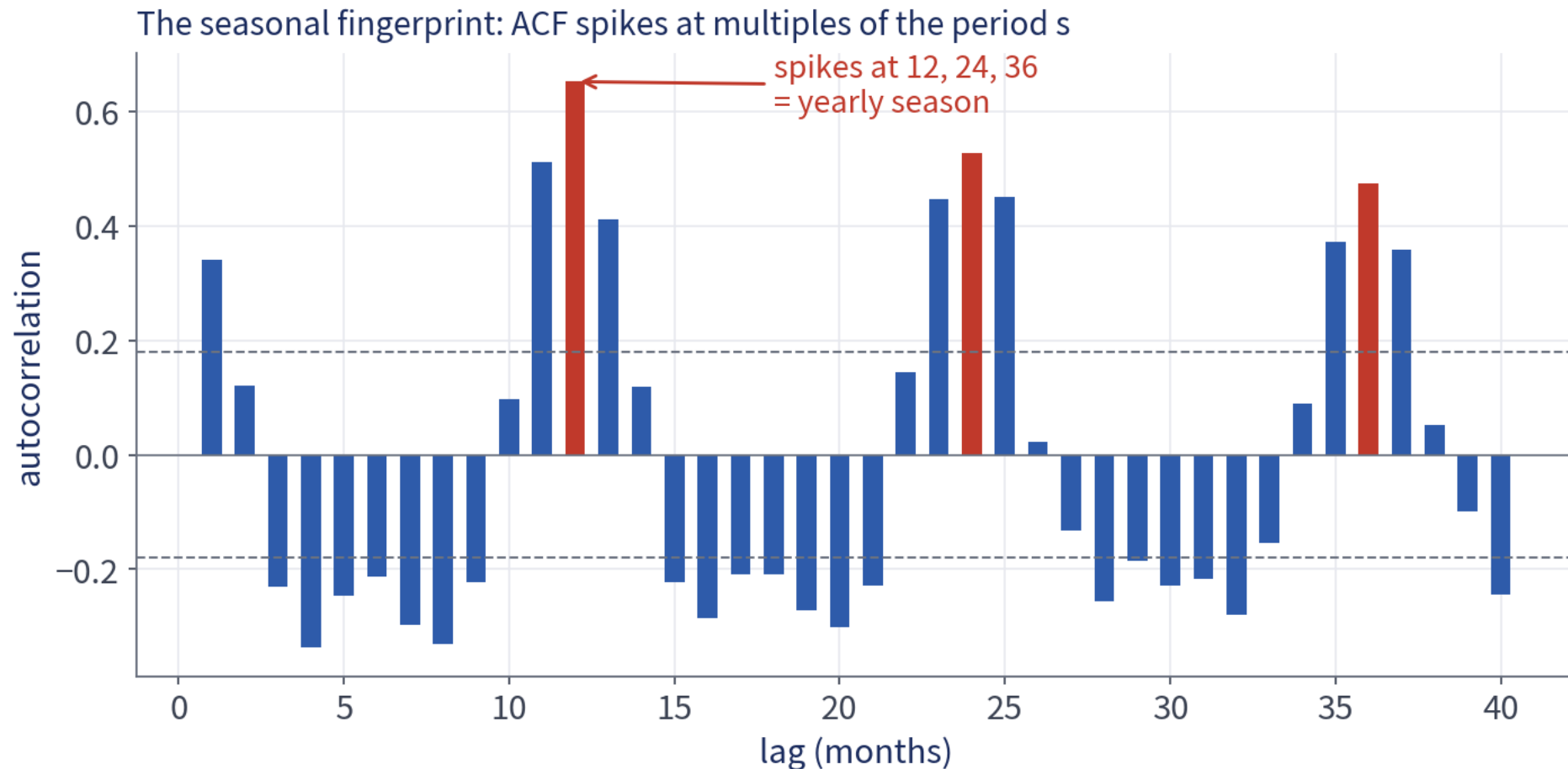
- Scan the ACF/PACF at lags s , $2s$, $3s$ specifically.
- A cut-off at the seasonal lags names the seasonal order.
- Slow decay across seasonal lags means you need a seasonal difference.

The two-region rule

- Near lags ($1, 2, 3\dots$) → the non-seasonal (p, d, q).
- Season lags ($s, 2s\dots$) → the seasonal (P, D, Q).
- Read the two regions independently.

The Seasonal Fingerprint

The ACF spikes at multiples of the period s (here 12, 24, 36). Those seasonal spikes are what tell you a seasonal term — or a seasonal difference — is needed.



SARIMA(p,d,q)(P,D,Q)_s

Two sets of orders, one at each timescale

$$\underbrace{\phi(L) \Phi(L^s)}_{\text{AR} \times \text{seasonal AR}} (1-L)^d (1-L^s)^D X_t = \underbrace{\theta(L) \Theta(L^s)}_{\text{MA} \times \text{seasonal MA}} \varepsilon_t$$

The two halves

- (p, d, q): the ordinary ARIMA at adjacent lags (Part 1).
- (P, D, Q): the SAME structure applied at the seasonal lag s.
- s is the period — 12 monthly, 4 quarterly, 7 daily.

Why it is elegant

- No new theory — the seasonal part mirrors the non-seasonal part.
- The two multiply together in one clean equation.
- Everything you learned in Weeks 3–4 transfers directly.

SARIMA, Decoded

Six integers: (p, d, q) handle the short-run dynamics and trend; (P, D, Q) handle the season at lag s . Same ideas, applied twice.

SARIMA, Decoded

Six integers: the first three handle ordinary dynamics, the last three handle the season.

(p, d, q) (P, D, Q) ^{s}
period

Non-seasonal part (p, d, q)

p - AR order on adjacent lags
 d - regular differences (the trend)
 q - MA order on adjacent shocks

Exactly the ARIMA of Part 1 -
short-run dynamics and the trend.

Seasonal part (P, D, Q) at lag s

P - AR order on seasonal lags $(s, 2s, \dots)$
 D - seasonal differences $(1 - L^s)$
 Q - MA order on seasonal lags

The same ideas, applied at the
season lag s (e.g. $s = 12$ for monthly).

The Seasonal Polynomials

AR and MA, written at the seasonal lag

$$\Phi(L^s) = 1 - \Phi_1 L^s - \dots - \Phi_P L^{Ps}, \quad \Theta(L^s) = 1 + \Theta_1 L^s + \dots + \Theta_Q L^{Qs}$$

Reading them

- $\Phi(L^s)$ is a seasonal AR polynomial in the lag operator L^s .
- $\Theta(L^s)$ is the seasonal MA counterpart.
- They act only at lags that are multiples of s .

The parallel

- These mirror $\phi(L)$ and $\theta(L)$ from the non-seasonal part.
- P and Q are the seasonal AR and MA orders.
- The full model is the product of both polynomials.

The Airline Model

The most famous SARIMA of all

$$(1 - L)(1 - L^{12}) X_t = (1 + \theta L)(1 + \Theta L^{12}) \varepsilon_t \quad \text{ARIMA}(0, 1, 1)(0, 1, 1)_{12}$$

What it is

- ARIMA(0,1,1)(0,1,1)_12 — one regular and one seasonal difference.
- Plus one MA term at lag 1 and one at lag 12.
- Just two parameters, yet it fits the airline data beautifully.

Why it endures

- It captures trend AND season with remarkable parsimony.
- It is the default starting point for monthly seasonal data.
- A perfect illustration of 'simple but useful' (Box).

Identifying the Seasonal Orders

A disciplined reading of the two lag regions

The routine

- First difference (d) and seasonally difference (D) as needed.
- Read the near lags for (p, q) ; read lags $s, 2s$ for (P, Q) .
- Keep all four orders small — parsimony still rules.

Practical tips

- Start from the airline model and adjust from its residuals.
- A single seasonal difference usually suffices ($D = 1$).
- Confirm with AIC/BIC and residual diagnostics, as always.

The Seasonal Cheat Sheet

Read each lag region separately

What you see	Near lags (1, 2, 3...)	Seasonal lags (s, 2s, 3s...)
ACF cuts off	→ MA order q	→ seasonal MA order Q
PACF cuts off	→ AR order p	→ seasonal AR order P
Slow decay	→ difference d	→ seasonal difference D

The whole trick: apply the Week-3 cheat sheet twice — once near lag 1, once around the season lag s.

Combining Regular and Seasonal Differences

How d and D work together

Trending, seasonal data often needs one of each: a regular difference for the trend and a seasonal difference for the pattern — applied together as $(1 - L)(1 - L^s)$.

The usual recipe

- $d = 1$ removes the trend; $D = 1$ removes the season.
- Together they flatten both at once.
- Higher d or D is rarely needed and risks over-differencing.

How to decide

- Difference for the trend first, then check for seasonal spikes.
- Add a seasonal difference only if those spikes persist.
- Stop as soon as the ACF is clean at both regions.

Building a SARIMA

The ARIMA loop, with a seasonal pass

1 · Difference

d and D

Regular difference for trend; seasonal for the pattern.

2 · Identify

both regions

Read near lags for (p,q); season lags for (P,Q).

3 · Estimate

fit + rank

Fit SARIMA; compare candidates by AIC/BIC.

4 · Diagnose

residuals

Check no seasonal spikes remain in the residuals.

5 · Forecast

season + trend

Predict; the forecast repeats the season.

Same five steps as Part 1 — you simply do the differencing and identification at two lags.

Forecasting With SARIMA

The forecast carries both the trend and the season

A SARIMA forecast reproduces everything it learned: the trend keeps rising and the seasonal shape repeats, cycle after cycle, with intervals that widen as the horizon grows.

What you see

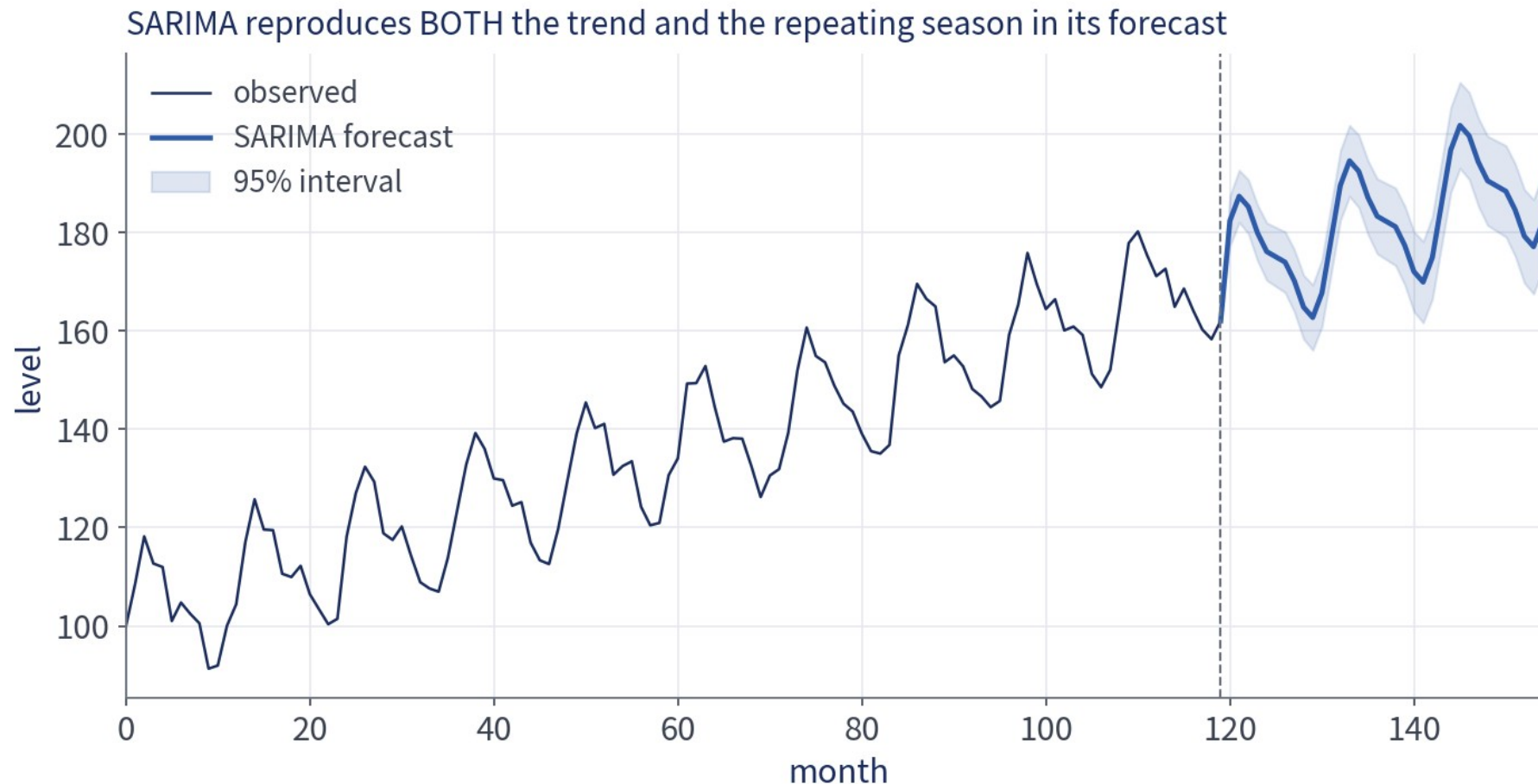
- The point forecast continues the trend (from the 'I').
- It overlays the repeating seasonal shape (from the 'S').
- The interval fans out, widest at the far horizon.

Why it is powerful

- One model captures level, trend, and season together.
- Short-horizon seasonal forecasts can be strikingly accurate.
- It remains a strong baseline for demand and macro data.

A SARIMA Forecast Repeats the Season

The forecast continues the upward trend while faithfully reproducing the yearly shape — trend from the 'I', season from the 'S', uncertainty from the fanning interval.



Seasonality in Finance

Powerful for some series, treacherous for returns

Where it genuinely helps

- Energy and commodity demand (heating in winter).
- Retail sales and volumes (holiday peaks).
- Quarterly macro series — GDP, earnings, inventories.

Where to be skeptical

- Raw return series show little robust seasonality.
- Calendar 'anomalies' are weak and often vanish out of sample (Week 1).
- Model seasonality on volumes and fundamentals, rarely on returns.

Fit a SARIMA to Seasonal Data

Direct Claude Code through the seasonal workflow

GOAL Take a clearly seasonal series, choose d and D , identify all six orders, and forecast the repeating pattern.

[INPUT] Prompt · Claude Code

Write and run `sarima_lab.py` that:

1. loads a monthly seasonal series (e.g. the classic airline-passengers data, $s = 12$);
2. plots it and its ACF/PACF, marking the spikes at lags 12, 24, 36;
3. applies a regular and a seasonal difference, then fits `SARIMA(0,1,1)(0,1,1)[12]` and one alternative, comparing AIC/BIC;
4. forecasts 24 months ahead with 95% bands and plots the level.

Explain which orders you chose and why.

How to work with it

- Look for ACF spikes exactly at 12, 24, 36.
- The forecast should repeat the yearly shape.
- Check residuals for leftover seasonal spikes.
- Keep it; Part 3 automates the whole search.

Deliverable: `sarima_lab.py` + a 24-month seasonal forecast with intervals.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1

Monthly data still spikes at lags 12 and 24 after one regular difference. What next?

Seasonal difference

Q2

In $\text{SARIMA}(p,d,q)(P,D,Q)_s$, which orders act at the seasonal lag?

(P, D, Q)

Q3

You fit $\text{ARIMA}(0,1,1)(0,1,1)_{12}$. What kind of data is this classic for?

**Monthly,
trend+season**

All three come straight from the two-region reading and the seasonal difference.

Three Pitfalls to Avoid

Where seasonal modeling goes wrong

Common mistakes

- "Raise the ARIMA order until it captures the season."
- "Seasonal spikes in the residuals are fine."
- "Returns must have a tradable seasonal pattern."

The reality

- Use a seasonal term at lag s — do not brute-force a high order.
- Leftover seasonal spikes mean the model is incomplete — revise.
- Return seasonality is weak and unstable; be skeptical (Week 1).

Three Things to Carry Forward

Part 2 in one screen

1

Seasonality is memory at the lag s

A pattern that repeats every s periods shows up as ACF spikes at $s, 2s, 3s$ — the same autocorrelation idea, moved out to the seasonal lag.

2

SARIMA is ARIMA applied twice

(p, d, q) at adjacent lags and (P, D, Q) at the seasonal lag s . A seasonal difference removes the pattern; seasonal AR/MA terms mop up the rest.

3

The airline model is your default

$\text{ARIMA}(0,1,1)(0,1,1)_{12}$ captures trend and season with just two parameters — the parsimonious starting point for monthly seasonal data.

Next — Part 3: building and forecasting ARIMA/SARIMA in practice, with automation and evaluation.